

joshua.f.allen@gmail.com 

GitHub 

Joshua Allen, PhD

Projects **The Football Factor Model**

- Modeled player performance based on values over replacement outcomes using Bayesian hierarchical models.
- Adjusted for contextual factors, including offensive and defensive EPA, rest differential, offensive/defensive efficiency, and age, using Gaussian processes.
- Model led to a 6-12 point increase in fantasy football scores per game.
- For a full writeup see [my website](#).

Football Power Rankings

- Built a Bayesian Bradley-Terry model to rank NFL teams by latent ability and incorporate uncertainty of the estimation.
- Designed a full-stack web app using FastAPI, React, and D3 for intuitive visualization of results.
- Translated ranking uncertainty into insights for consumer preference modeling and competitive benchmarking. [Live Demo](#)

Memorial to The Deported Jews of France

- Built a geospatial data pipeline to collect and geocode 70,000 addresses using OSM and other free APIs.
- Built reproducible data pipelines and R Shiny applications for public reporting and policy-facing dissemination
- Used R Shiny and Mapbox GL to show the distribution of deportations across France.
- For a demo see [this repository](#)

Experience **Georgia State University** **Computational Social Scientist**

Aug. 2019 - July 2025

- Designed and executed a quasi-experimental study using Double Machine Learning (Bayesian Additive Regression Trees), Synthetic Control, and Sensitivity tests to evaluate long-term political outcomes.
- Developed and led a multi-year research project from inception to completion, resulting in a new spatial understanding of the Holocaust using NLP and LLMs.
- Implemented and managed end-to-end data pipelines covering all phases of the data lifecycle using R and Quarto. Pipeline **improved data deliverable time from 3 days to less than 2 hours**.
- Implemented reproducible and scalable Python workflows to collect, ingest, and clean 3 terabytes of data locally for natural language processing.

- Led the development of the **ecdata R and Python packages** to deliver the largest open-source political executive communication text dataset.
- Developed 18+ production-grade web scrapers in R and Python for large-scale data collection, resulting in the largest open-source dataset of world leaders' communications.
- Mentored junior data scientists and advised colleagues on statistical analysis, scientific programming, and best practices for reproducible research.
- Leveraged data simulations and data visualization to communicate complex statistical concepts and social science concepts to non-technical stakeholders.

Invited Talks

- Southern Political Science Association Annual Meeting: 2025, 2024, 2020
- International Studies Association Annual Meeting: 2023, 2022, 2019
- Atlanta Symposium on Political Science: 2020, 2019

Skills Languages: SQL, Python, R, Stan, Stata
 Modeling: PyMC, Brms, Scikit-learn, Tidymodels
 Data Engineering: Docker, Git, PySpark, Polars, Pandas
 Frontend Development: React.js, D3.js, RShiny
 Causal Inference, Bayesian Statistics, Machine Learning, and Written & Verbal Communication

Education **Ph.D** Political Science (Comparative Politics and Research Methods): Georgia State University (2019 - 2025)
Masters of Arts, Political Science: Georgia State University (2017 - 2019)
Bachelors of Arts, Political Science: Sonoma State University (2013 - 2017)